

Becoming an AI Research Scientist: Tasks, Skills, and Learning Path

AI research scientists at leading labs (Google Brain/DeepMind, Microsoft, Meta, OpenAI, etc.) focus on *inventing* new models and algorithms – not just using existing AI tools. Their daily work is heavy on **literature review and experimentation**, algorithm design, and large-scale engineering. For example, they spend much of their time reading new papers, **analyzing data and prototyping** novel models ¹ ². In practice this means they *write and test code to develop new neural architectures*, train models on custom data pipelines, and measure performance on research benchmarks ³ ². Their projects often involve running massive experiments (e.g. training multi-billion-parameter transformers) and collaborating across teams, then publishing the findings. In sum, AI researchers typically **(a)** carve out a problem (e.g. “better few-shot learning for language”), **(b)** survey and digest hundreds of related papers daily ⁴, **(c)** design and code up new model variants or training methods, and **(d)** run large experiments to validate ideas, often on huge GPU/TPU clusters. They also **share results** via internal reviews and external publications ¹ ³.

- **Core tasks:** Read and research (study latest ML/AI papers, explore new ideas); design algorithms and models; write “research code” (prototypes, not production code) to **train and test** those models; iterate and optimize based on results ¹ ². This is unlike a typical data scientist who might primarily apply off-the-shelf models – research scientists build or modify the models themselves.
- **Collaboration & Publication:** They work closely with other scientists and engineers (often in interdisciplinary teams) to refine ideas, and they publish papers or reports to contribute to the field ⁵ ². Publications and open-source releases (e.g. model code, data sets) are a key part of the job.
- **Cutting-edge development:** For companies building *large language models (LLMs)* or other foundation models, research groups handle *everything* from data to deployment. This includes curating vast training corpora, engineering specialized tokenization schemes, designing novel Transformer variants, and devising new training methods ⁶ ⁷. For example, Google’s PaLM model (540B parameters) was trained on thousands of TPU chips with a custom “Pathways” distributed system, achieving record efficiency by redesigning the Transformer computation flow ⁸ ⁷. In short, they manage **large-scale pipeline engineering**: parallelizing training (data/model/pipeline/tensor parallelism) across GPU/TPU clusters ⁹ ⁸, handling data cleaning and tokenization at petabyte scale ⁶, and constantly tweaking hyperparameters and architectures to push performance.

In practice, research teams building LLMs will: curate and preprocess huge multi-language text (and sometimes code/image) datasets; experiment with model **architectures** (e.g. Transformers, sparse MoE layers, retrieval-augmented models); implement **distributed training** (using strategies like data/pipeline/tensor parallelism as described by OpenAI) ⁹; and evaluate models on benchmarks (or novel tests). They also investigate model behavior (e.g. language reasoning, factual accuracy) and safety/alignment issues. In contrast to an ML engineer who might “use BERT” or “run Hugging Face,” these scientists often build custom training pipelines or even new ML frameworks to scale up models. (Google’s Pathways system is one example of an in-house training framework ⁸.)

Skills and Learning Path

Becoming an AI research scientist requires **deep technical preparation** and usually advanced study. A strong foundation in mathematics (linear algebra, probability/statistics, optimization) and computer science is essential ¹⁰ ¹¹. Many research roles (especially at top labs) expect a Master's or PhD in a quantitative field (CS, EE, physics, math, etc.) because this training develops the required depth ¹² ¹³. However, the key is *research ability*: building a track record of understanding and extending ML ideas. As one guide notes, AI research scientists often “will almost definitely need a PhD and a deep specialism in machine learning” areas like NLP, vision or RL ¹².

Key skills to develop: In addition to math, you should master programming (especially Python and C++/CUDA for performance) and learn major ML frameworks (PyTorch, TensorFlow, JAX). As Teerapong Panboonyuen (AI researcher) advises, **“gain proficiency in tools like Python and PyTorch, and dive into machine learning by writing algorithms from scratch”** ¹¹. Practical experience is crucial: implement neural network components by hand (e.g. build a Transformer block from basic operations) and train small models end-to-end. Participate in challenges or open projects: for example, Kaggle competitions and other hackathons are useful to practice ML engineering ¹⁴. Contributing to open-source ML code (e.g. on GitHub or Hugging Face libraries) can also demonstrate your skills.

Study and practice: Start with solid machine learning fundamentals (online courses, textbooks). The Coursera career guide emphasizes learning ML/DL concepts deeply. Build simple projects (classification, regression, small NLP models) to cement understanding. Then *increase scale and complexity*: train larger models (e.g. fine-tune open LLMs, experiment with GPUs), and learn about data pipelines and training at scale. The AI researcher's roadmap recommends writing ML algorithms from scratch and deploying models (e.g. using Streamlit or similar) to gain end-to-end insight ¹⁵. Studying existing papers is also a hands-on exercise: try to re-implement key models (many open-source implementations exist for popular papers) to grasp the details.

Research mindset: A daily habit of reading new research is indispensable. As one AI researcher puts it: *“Each day, I read at least one hard paper in detail, and I quickly scan a handful of easier papers... I talk to at least one researcher in another area to cross-pollinate ideas.”* ⁴. Regularly follow top conferences (NeurIPS, ICML, ICLR, ACL, etc.) and journals, and subscribe to arXiv notifications in your area. Writing blog posts or technical notes on new papers can sharpen your understanding. Engage with the research community (forums like [Hugging Face forums](#), ML Twitter, research seminars) to stay current ¹⁶ ⁴.

Education and experience path: Formally, an advanced degree (MS/PhD) in ML/AI is a direct route, because it provides research experience. Many researchers do a PhD focusing on an ML problem, then transition into industry R&D. If a PhD isn't feasible, you can still enter research via “research engineer” or “applied scientist” roles by demonstrating your skills: contribute to research projects at your company, publish or present findings, or collaborate with academics. Online specializations (e.g. Fast.ai, MIT/HuggingFace NLP courses) and self-study can help build credentials, but nothing substitutes real research work. internships at AI labs or competitions can also lead to opportunities.

Continuous learning: AI is rapidly evolving, so commit to lifelong learning. Upgrade your knowledge in deep learning (understand concepts like attention, backpropagation, advanced optimizers), and explore specialized topics (transformers, meta-learning, reinforcement learning, etc.). Resources like distill.pub, cs231n lectures, or Andrej Karpathy's “Neural Networks: Zero to Hero” can deepen intuition. Hands-on

projects should keep advancing: try building small LLMs (there are tutorials for training mini-Transformers) and explore modern toolkits (e.g. JAX/Flax, DeepSpeed, TensorRT). The roadmap by Panboonyuen also suggests studying fine-tuning and retrieval-augmented generation (RAG) techniques once you are comfortable with basics ¹⁴ ⁶ .

In summary, AI research scientists are essentially developers of *new* AI. They bridge theory and engineering by inventing models and training them at scale. To prepare, leverage your strong math/ML background by diving into research: read and implement papers, practice large-scale training, and if possible pursue a research-focused degree or project. Build a portfolio (code repositories, papers, blog posts) that shows you can tackle novel AI problems. With persistence and deep technical learning, you can follow in their footsteps.

Sources: Industry and career guides on AI research roles ¹ ¹⁷ , personal accounts by AI researchers ⁴ ² , and technical write-ups of large-model training ⁸ ⁹ were used to compile this overview. These highlight the responsibilities of AI research scientists (algorithm design, experimental development, publication) and the advanced skills they apply when building LLMs. The learning path is drawn from career advice blogs ¹³ ¹¹ and educational resources. Each recommended step is supported by the cited sources.

¹ ³ ⁵ What Is an AI Research Scientist? 2025 Career Guide | Coursera

<https://www.coursera.org/articles/ai-research-scientist>

² ¹¹ ¹⁴ ¹⁵ ¹⁶ Career Paths for AI Research Scientists: My Journey and Insights | Teerapong Panboonyuen

<https://kaopanboonyuen.github.io/blog/2024-09-01-career-paths-for-ai-research-scientist/>

⁴ A day in the life of an AI Researcher | by Rik Koncel-Kedziorski | Kensho Blog

<https://blog.kensho.com/a-day-in-the-life-of-an-ai-researcher-e5c3a563fd2d?gi=c88129351816>

⁶ ⁷ ⁸ Pathways Language Model (PaLM): Scaling to 540 Billion Parameters for Breakthrou

<https://research.google/blog/pathways-language-model-palm-scaling-to-540-billion-parameters-for-breakthrough-performance/>

⁹ Techniques for training large neural networks | OpenAI

<https://openai.com/index/techniques-for-training-large-neural-networks/>

¹⁰ ¹³ ¹⁷ Refonte Learning : How to Become AI Research Scientist in 2025

<https://www.refontelearning.com/blog/how-to-become-ai-research-scientist-in-2025>

¹² Do you need a PhD for Machine Learning? | by Egor Howell | Data Science Collective | Jun, 2025 | Medium

<https://medium.com/data-science-collective/do-you-need-a-phd-for-machine-learning-7b9c0330d22f>